Memory managment

*Deff: memory managment is the function resposible for managing the computer's primary memory.

$\boxed{\text{cpu}} \rightarrow \boxed{\text{memory}}$

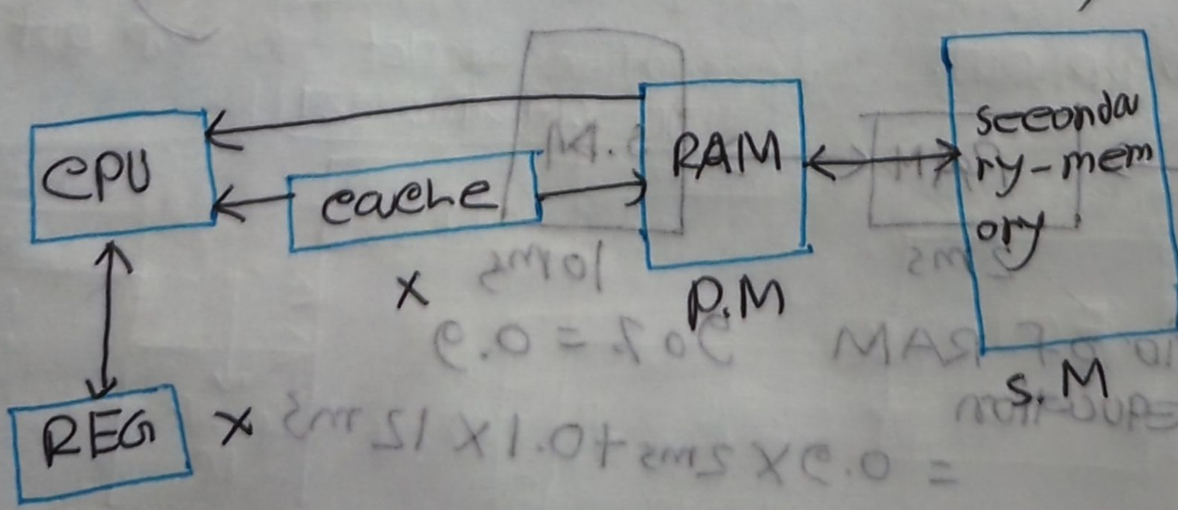
process   program

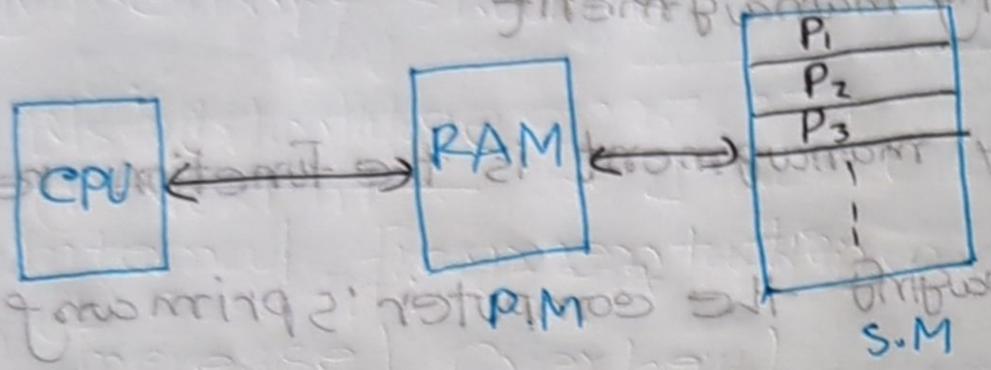
* process কে cpu exequet করে
* memory তে program থাকে

** Memory requirement

i) Size — memory size
ii) Accesess time → exequt time
iii) Per unit cost → ~~se~~ less cost

} Not posseble for a memory



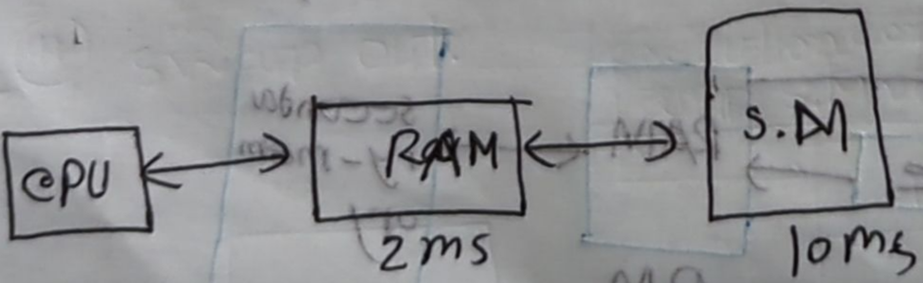


* Cpu directly কোনো process কে S.M থেকে exequte করতে পারবেনা.

CPU ⟶ logical Address

⟶ Phycical address

* calculate total access time of memory



Hit ratio of RAM 90% = 0.9

equation

$= 0.9 \times 2ms + 0.1 \times 12ms$

$= 1.8 + 1.2$

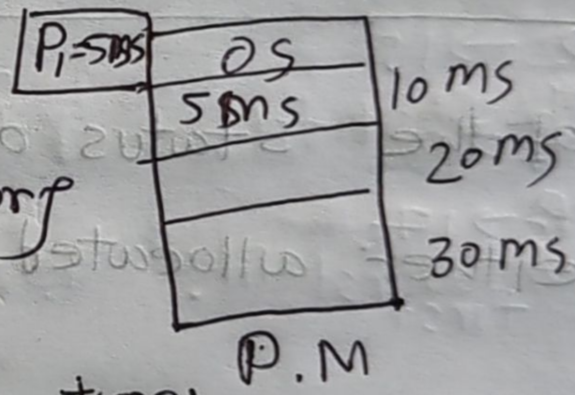
$= 3ms$

Q. * what is locality of Reference is a Phenomenon in which a computer program tends to access same set of memory locations for a particular time period.

Q. memory Allocation type.

* Contiguous memory allocation (CO4 ~~[illegible]~~ [illegible]

Is a memory allocation method that allocates a single contiguous section of memory to process



* Advantage
① Simplicity
② Efficiency
③ low fragmentation
④ First access

type:
① Fixed size partition
② variable size partition

* disadvantage ① Limited flexibility ② memory wastage
③ Extranal fragmentation.

ভাংগা জাগা করে

* Non contiguous : is a memory allocation method that allocates separate Blocks of memory to a process

* Advantage:
① Reduced External fragmentation
② Increased memory utilization
③ Flexibility.

(IV) memory sharing

* Disadvantage

(i) Internal fragmentation

(ii) Increased Overhead

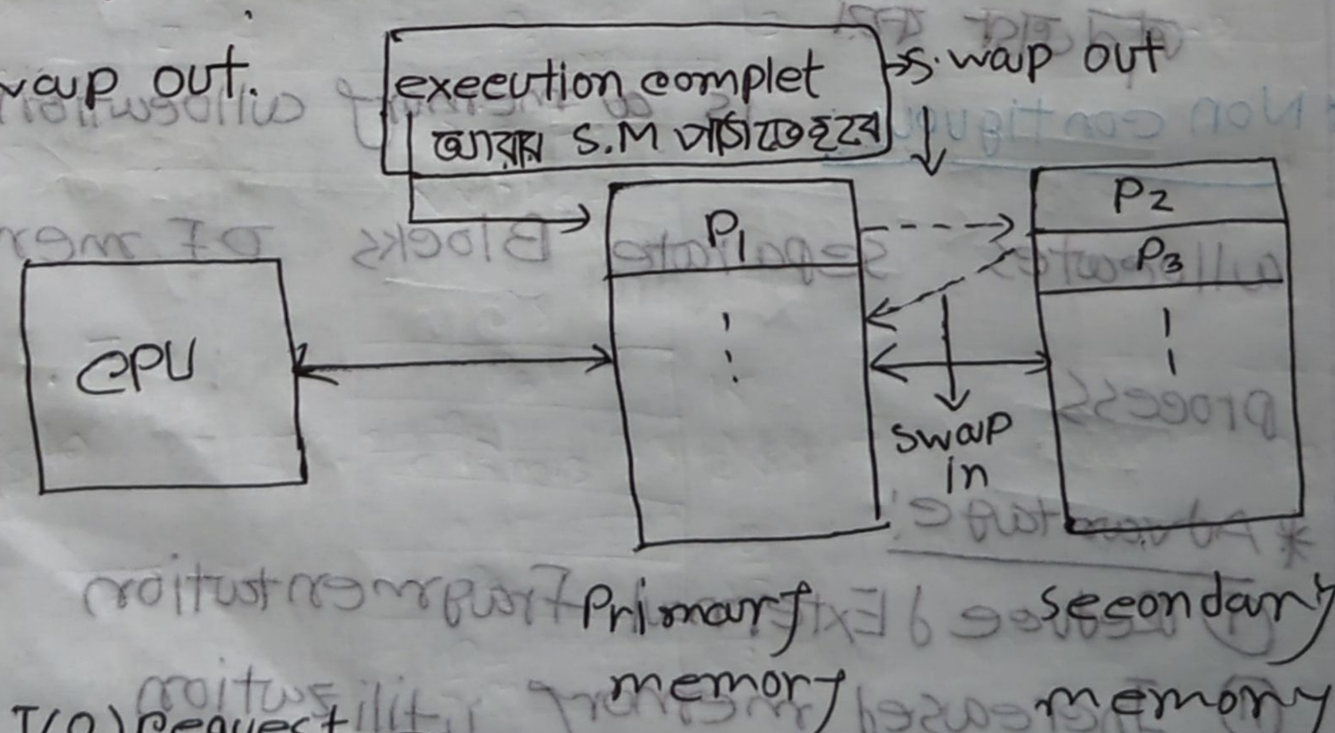
(iii) slower access .

Memory management function that Keeps tracks of the status of each memory location either allocated or free.

* Swapping in Memory managment

(i) swap in

(ii) swap out.



* P1 (I/O) request আসলে তাহলেও swap out হবে

Fragmentation: Primary memory কত টুকু জায়গা wast হচ্ছে সেটা

① **Internal Fragmentation**: occurs when allocated memory is larger then the requested memory, Leaving unused space within the allocated block.

② **External fragmentation**: occurs when free memory is divided into small, non contiguous blocks making it diffieult to allocate large blocks of memory, even though there is enough total free memory

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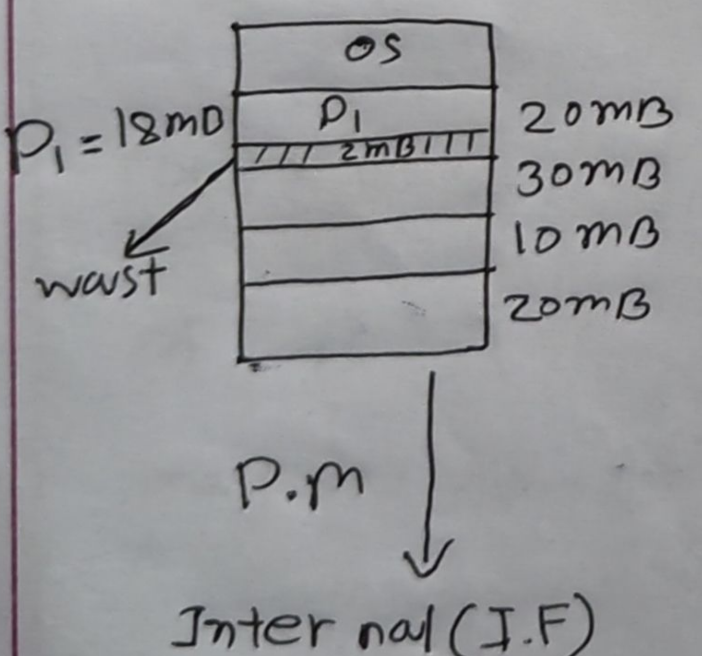
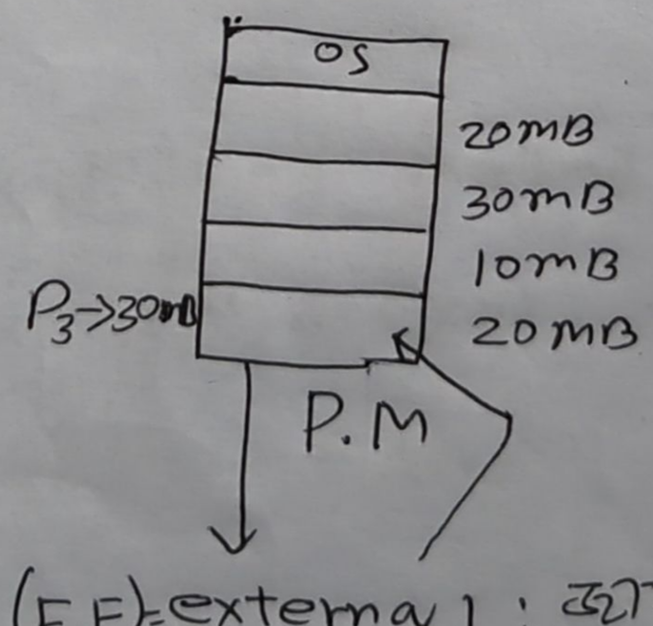
for contiguous

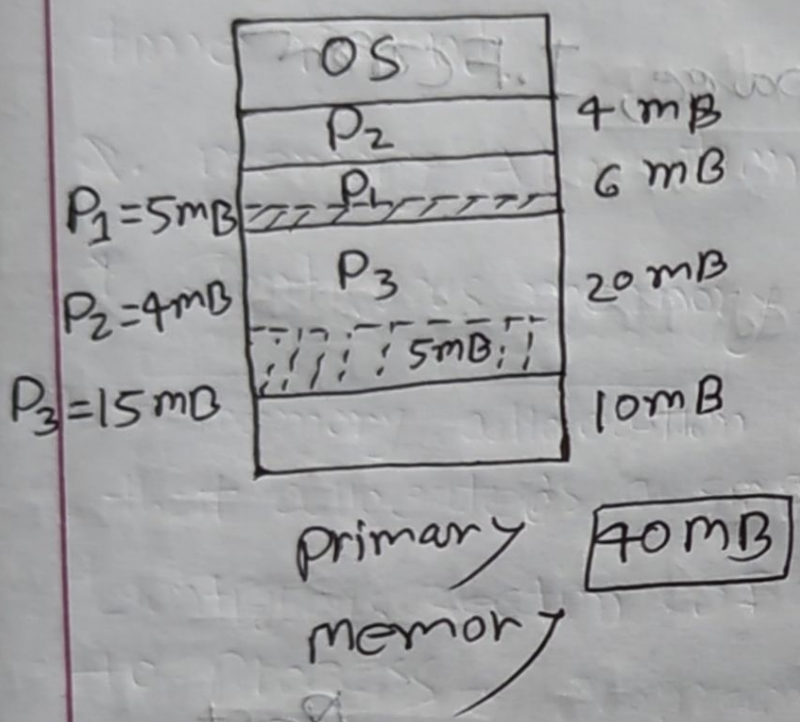


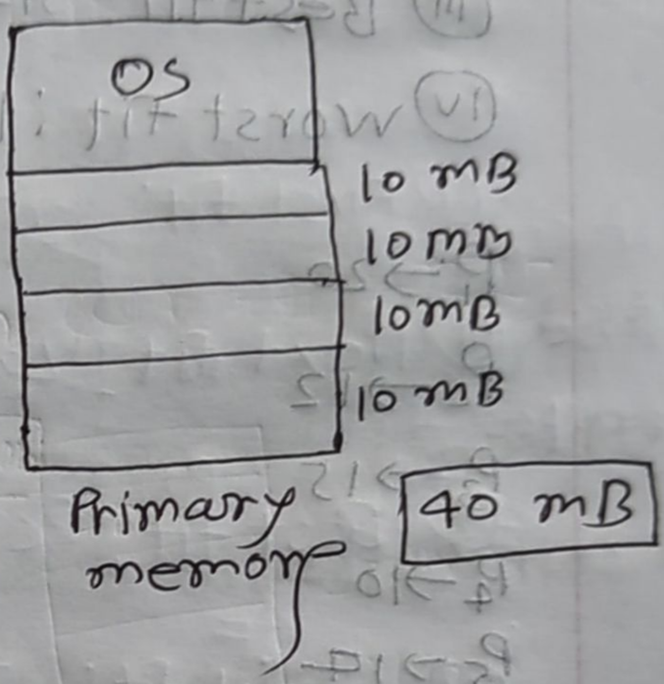
Internal (I.F)



(E.F)-external : জায়গা থাকার পরেও কোনো process Allocate না করতে পারা

Fixed Size Partioning Contiguous Memory Allocation





Problem: Internal fragmentation or memory wast when the process size is smaller then the capacity

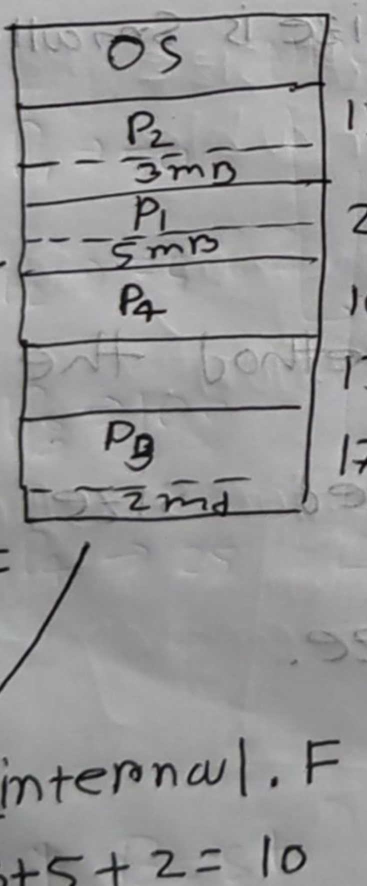
Fixed size: In this partitioning method the numder of partition are all a fixed size, but they may or may not be same size.

Algorithm

(I) First fit – প্রথম থেকে search
(II) Next fit : শুরু প্রথমথেকে তারপর Allocate process থেকে শুরু হবে
(III) Best fit : সবচেয়ে ছোট space হতে হবে, I.F কম
(IV) worst fit : largest space, I.F বেশি

$P_1 \to 20$
$P_2 \to 12$
$P_3 \to 15$
$P_4 \to 10$
$P_5 \to 17$

First

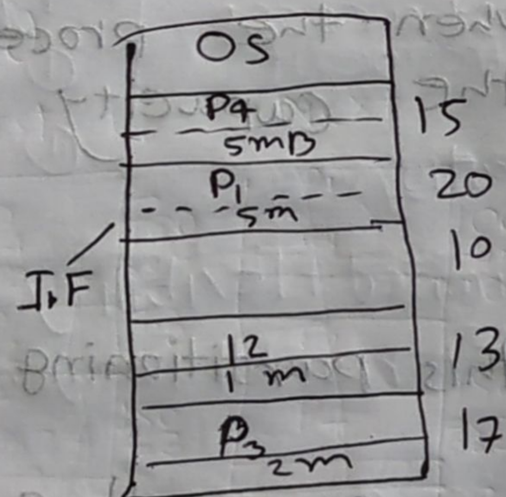


I.F

P5 = E.F

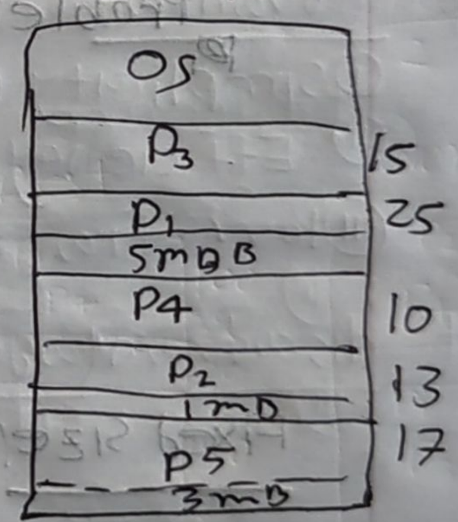
calculate internal. F
3+5+2 = 10

Next

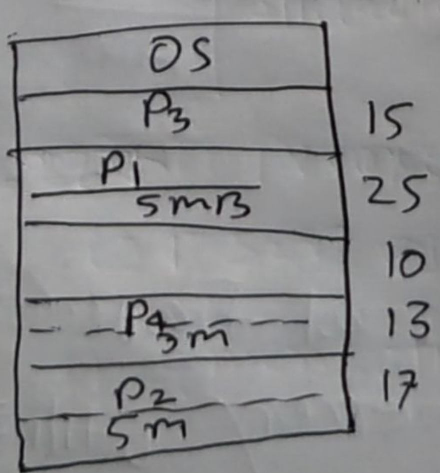


P5 = E.F

Best



worst

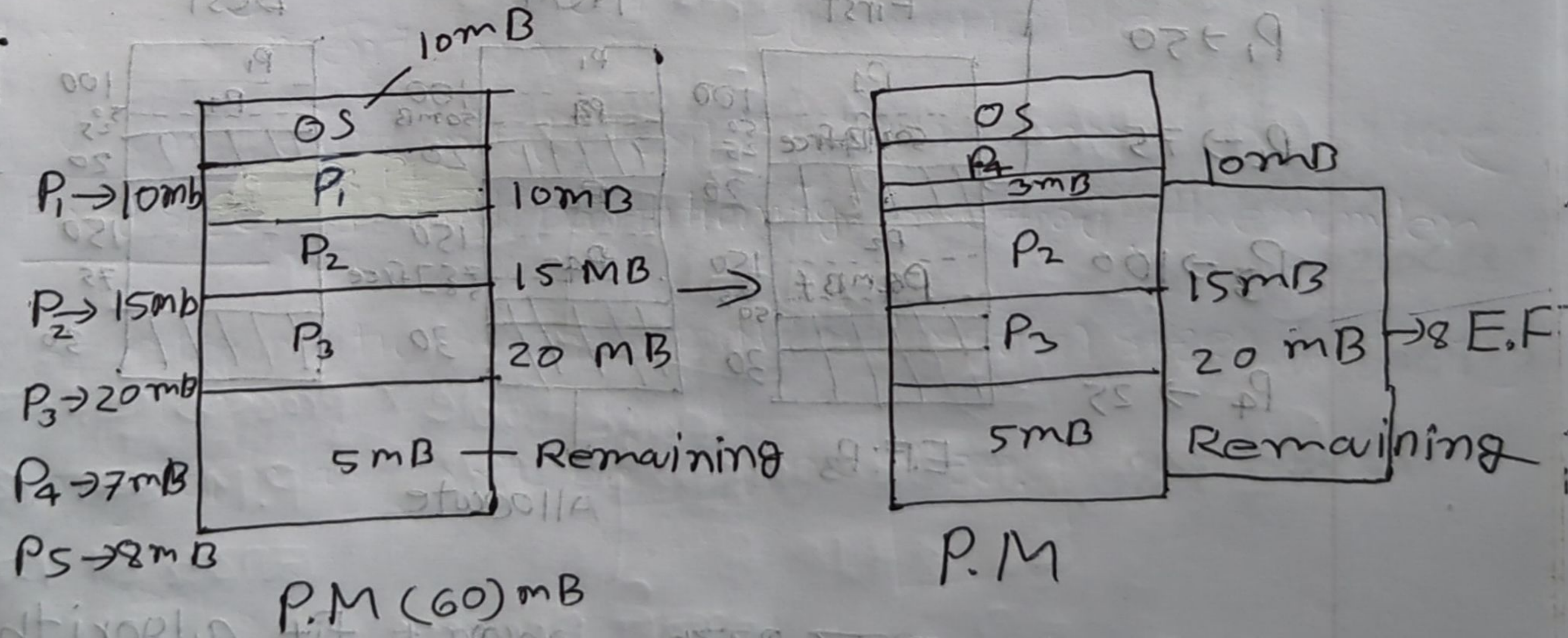


* Best fit Algorithm Fixed size partition এর জন্য ভালো কারণ I.F কমে।

variable size partitioning

contiguous Memory Allocation

variable size: This partition the memory is divided into section, where the size of each partition is determined besed on the size of the process being ~~executed~~ loaded into memory.



* P_1 exeqution complete হয়ে যায় তবে Partition আবার free আমরা নতুন Process ~~execute~~ execut করতে পারব।

E.F solved method: compaction (Process swaping)

Algorithm

(i) First fit: প্রথম থেকে search

(ii) Next fit: Last allocated Process থেকে

(iii) Best fit: ছোট space

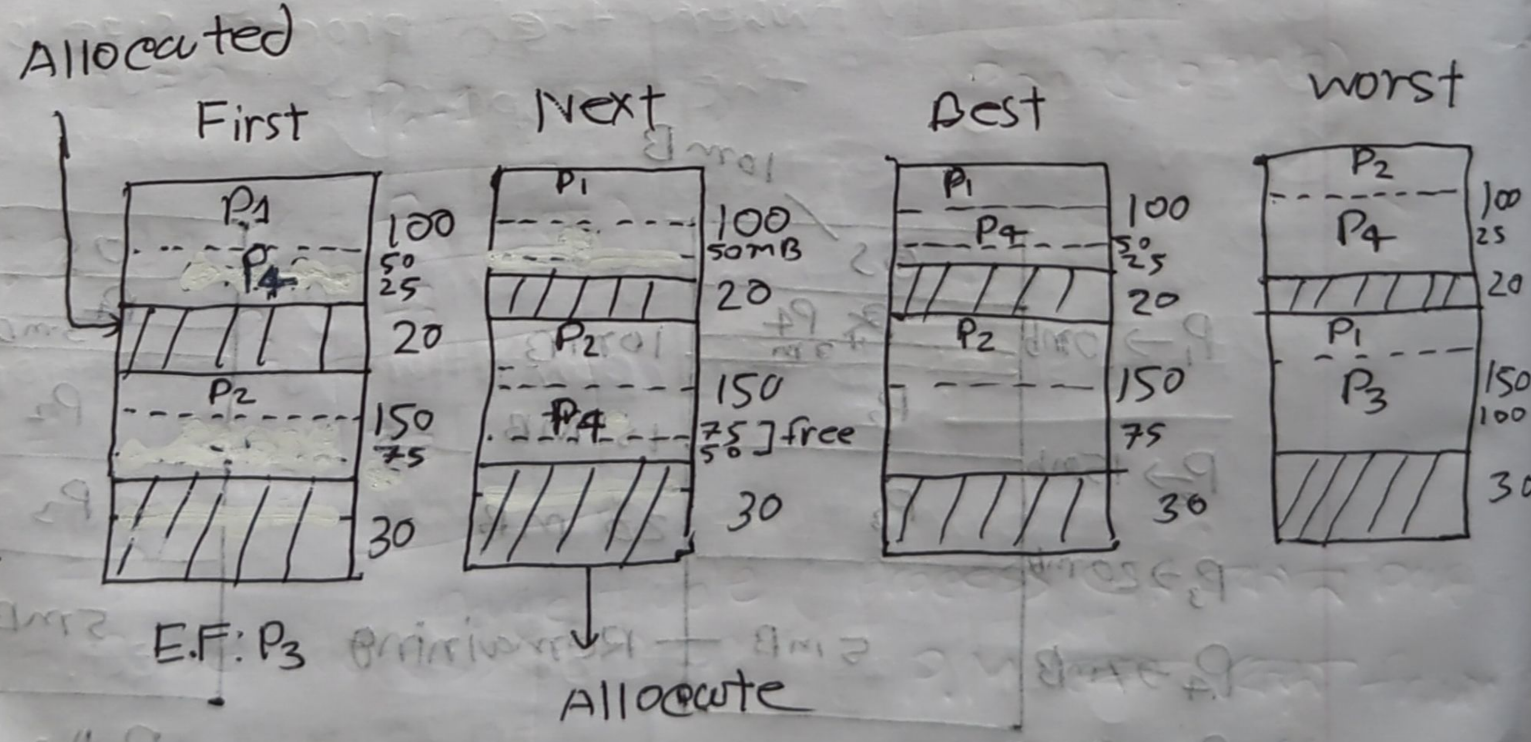
(iv) worst fit: বড় space

$P_1 \to 50$

$P_2 \to 75$

$P_3 \to 100$

$P_4 \to 25$



* Best এর ক্ষেত্রে worst fit algorithm কারণ

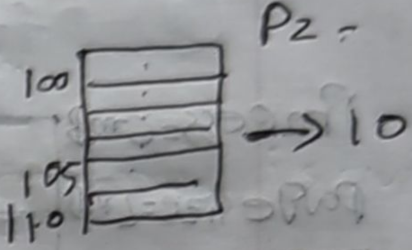
I.F হয়

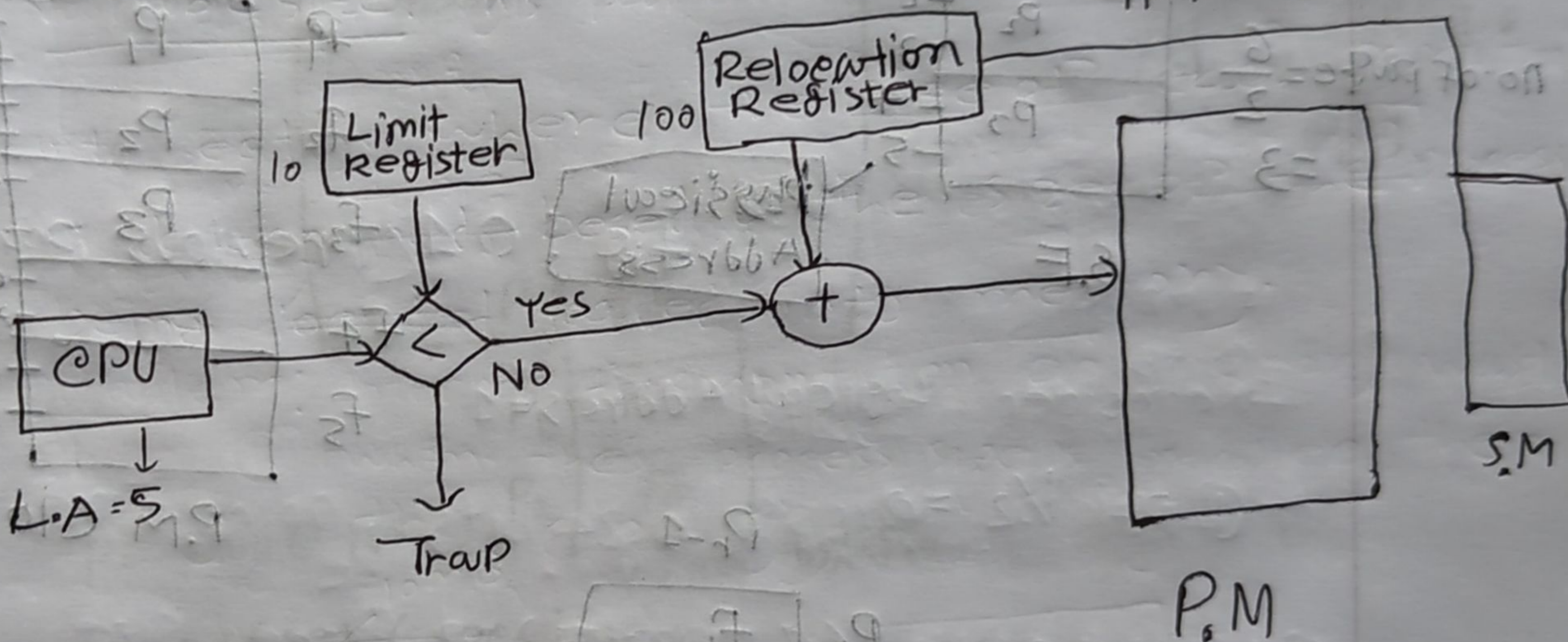
Address Translation for contiguous memory Allocation

Limit Register: Stores the size of process

Relocation Register: store the base address of the process.

Physical Address: Base Address + Logical Address
100 + 5 = 105





Process	Limit.R	Relocation R	logical Addres
P_0	380	700	210 → 210 < 380
P_1	425	1100	500 → 500 < 425
P_2	115	200	80 → 80 < 115

Physical Addres

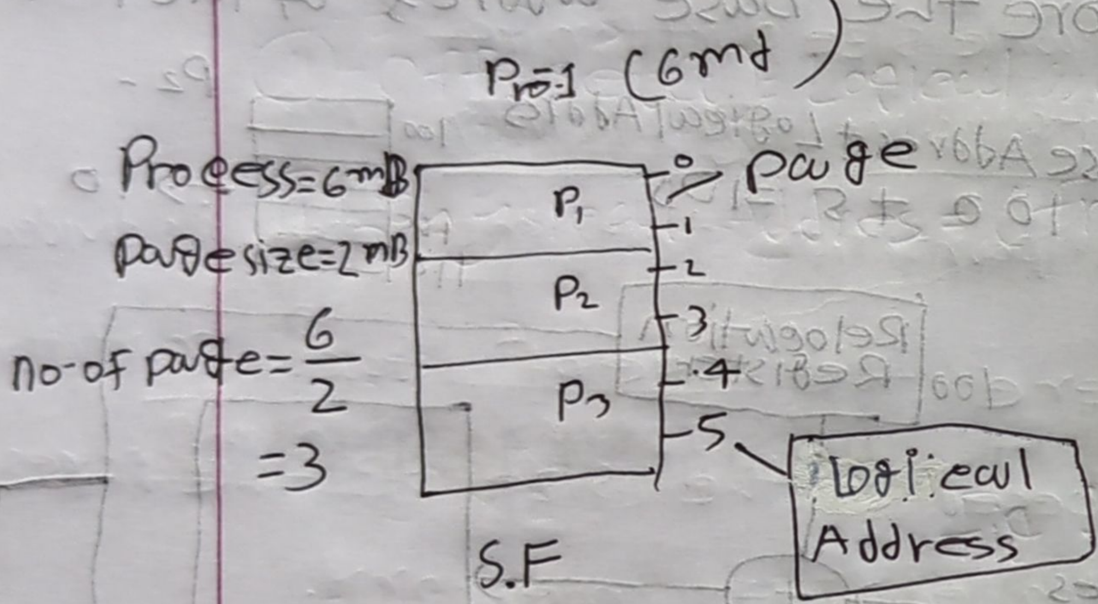
condition true -

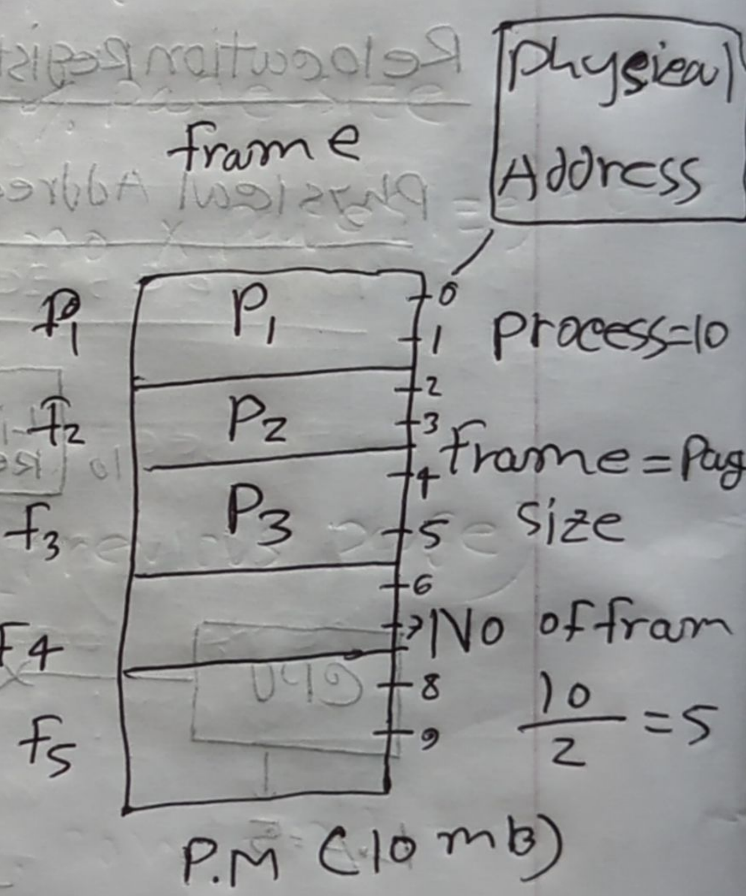
700 + 210 = 910	P_0
Trap (False)	P_1
200 + 80 = 280	P_2

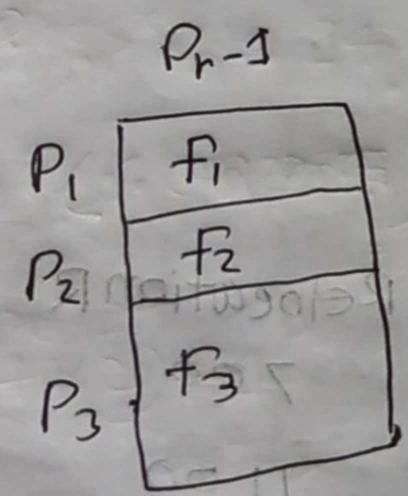
Paging

Non contiguous memory

* Proces ভেঙে ফেলব।







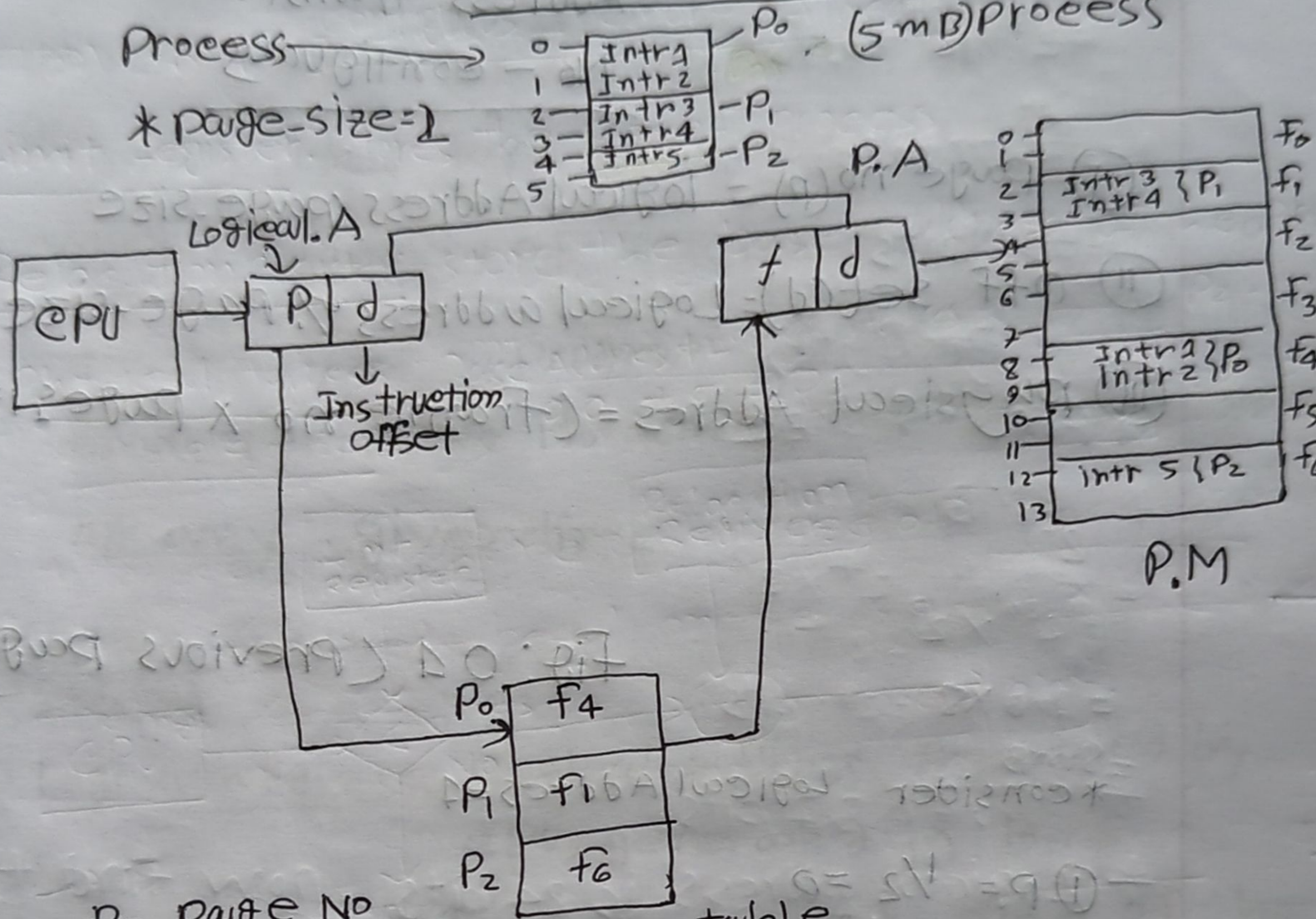
Page table (Process এর কোন page P.M ~~এর~~ কোন frame আছে তা নির্ধারণ করে)

প্রতিটি process এর জন্য প্রয়োজন

* Internal fragmentation আছে

Fig:01

Paging Architecture



page table

P= page No

d= offset (location)

f= frame No

Intr= Instruction

* CPU Logical Address generate করে process এর জন্য

* logicalA =1 হলে ~~করে~~ তার উত্তর P.A=9

*

Address translation

No – contiguous

① page no(p) = logical Address / page_size

⑪ Off set(d) = Logical address % page-size

⑪ physical Addres = (frame no x pagesize) + offset

Fig: 01 (previous page)

*consider Logical Address = 1

① $p = 1/2 = 0$

⑪ $d = 1 \% 2 = 1$ (% ভাগশেষ)

⑪ $P.A = (4 \times 2) + 1$

$= 9$

Ⓘⓥ $P.A$ = Base Address of frame + Offset

$= 8 + 1$

$= 9$

Q. Different between page and frame;

(I) Page used for logical address.

(II) fram used for Physical address.

Q. what is locality of Referance: ~~P.~~ rafers to the tendency of a program to access to relatively small portion of it's address space ~~frequency~~ at any given time.

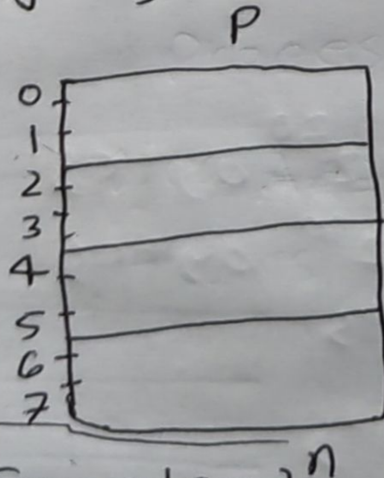
$\boxed{\text{* math}}$

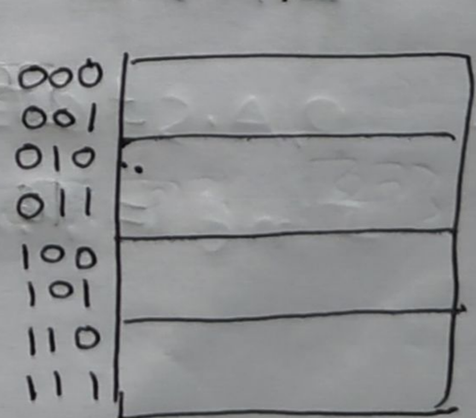
(I) Logical Address: Logical address generated by CPU, generated in Bits

(II) Logical Address: Space: Total size of the process; represent by bytes.

* ~~Frame~~ process size = 8 byts

* page size = 2 byts





2^{10} bytes = 1 KB
2^{20} " = 1 MB
2^{30} byte = 1 GB
2^{40} " = 1 TB

bit formula $= 2^n$
$= 2^3 = 8$

* Find 8 mB

$= 2^3 \times 2^{20} \rightarrow$ byts

$= 2^{23} \rightarrow$ bits

$= 23$ bits

* process size

2^{23}

$= 2^{20} \times 2^3$

$= 1mB \times 8$

$= 8mB$

* Logicau address is 24 bits, now find the Logical Addres space or Procees size

$L.A = \cancel{20}\ 24$ bits

$L.A.S = 2^{24}$ bytes

$= 2^{20} \times 2^4$

$= 16MB$

* ~~LogLA~~

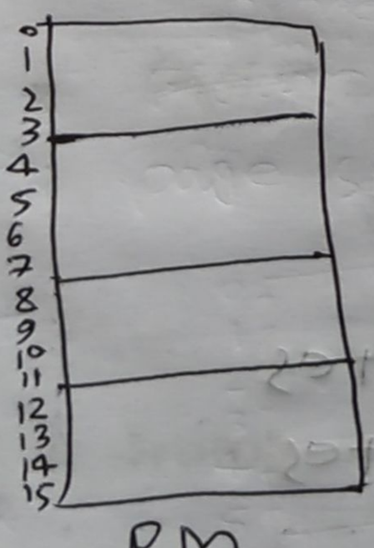
L.A.S = 8 GB find L.A

$$L.A = 2^3 \times 2^{30} \text{ bytes}$$
$$= 2^{33}$$
$$= 33 \text{ bits}$$

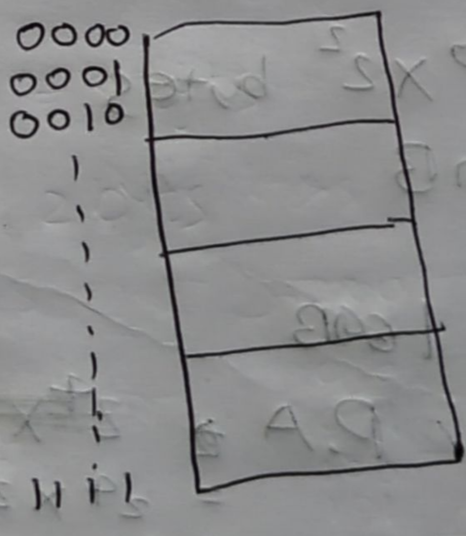
(I) <u>Physical address</u>: The address of the ~~P~~ primary memory represent in bits

(II) <u>P.A space</u>: Total size of ~~process re~~ P.M, represent in bytes

P.M size = 16 bytes
From size = 4 bytes



P.M



2^{10} bytes	= 1K B
2^{20} "	= 1M B
2^{30} "	= 1G B
2^{40} "	= 1T B

* $2^n \rightarrow 16$
2^4 bits

* total memory size

$2^{4 \text{ bits}}$

$= 16$ ~~Bits~~ Bytes

* 16MB

$= \quad 2^4 \times 2^{20}$
$= \quad = 2^{24}$
$= 2^4$ bits

* $P.A = 32$ bits

find P.A.S or total P.M size

2^{32} bits

$= 2^{30} \times 2^2$ bytes
$= 4GB$

* P.A.S 16GB

Find : P.A $=$ ~~$2^4 \times 2^{30}$ bytes~~
$= 2^4 \times 2^{30}$ bytes
$= 2^{34}$ bits

$= 34$ bits

P.S = 33

P.A. = 24

page size 2^{11} KB = frame size

No of page = ?

" " frame = ?

* Memory size = $2^{30} \times 2^{3}$

= 8 GB

* $= 2^{20} \times 2^{2}$

$= 4$ MB

*

22 P	d

$\longleftarrow 33 \longrightarrow$

$(33 - 11) = 22$

13 F	d

$\longleftarrow 24 \longrightarrow$

$(24 - 11) = 13$

* page $= 2^{22}$

$= 2^{20} \times 2^{2}$

$= 4$M

* fram $= 2^{13}$

$= 2^{10} \times 2^{3}$

$= 8$K